

Lawrence R. Liu

(650) 796-4234 | lawrencerliu@ucla.edu |   

EDUCATION

University of California Los Angeles

MS, Electrical Engineering, GPA: 4.0/4.0

Los Angeles, CA

Sep 2024 – June 2026

University of California Los Angeles

BS, Electrical Engineering, GPA: 3.8/4.0

Los Angeles, CA

Sep. 2021 – June 2024

RESEARCH EXPERIENCE

Graduate/Undergraduate Researcher

April 2023 – Present

Advised By Professor Lin F. Yang

UCLA

- Developed new novel LLM compression algorithms for efficient LLM inference
- Achieved competitive performance against SOTA pruning and quantization algorithms with 48x calibration data compared with SOTA VQ methods
- Published in COLM and ICLR workshops

Undergraduate Researcher

February 2022 – October 2024

Advised By Professor Vwani Roychowdhury

UCLA

- Trained and pruned a lightweight deep learning classifier for epilepsy biomarker detection
- Developed unsupervised time-frequency VAEs and graph-based models in PyTorch to discover more accurate, faster to detect, biomarkers for predicting surgery outcomes
- Led a multidisciplinary team to build a Python GUI tool for real-time brain signal recording and biomarker detection, now actively used by pediatric neurologists at UCLA

Publications

- **Lawrence Ray Liu**, Inesh Chakrabarti, Yixiao Li, Mengdi Wang, Tuo Zhao, Lin F. Yang. “NoWag: A Unified Framework for Shape Preserving Compression of Large Language Models”, accepted in *COLM 2025*
 - * Preliminary results presented at ICLR 2025 Sparsity in LLMs Workshop
- Yipeng Zhang, Atsuro Daida, **Lawrence Liu**, et al. “Self-Supervised Data-Driven Approach Defines Pathological High-Frequency Oscillations in Epilepsy”, accepted in *Epilepsia*, 2025.
- Yipeng Zhang, **Lawrence Liu**, Yuanyi Ding, et al. “PyHFO: Lightweight Deep Learning-powered End-to-End High-Frequency Oscillations Analysis Application”, In *Journal of Neural Engineering*, 21, 2024.
- Tonmoy Monsoor, Atsuro Daida, Prateik Sinha, Yipeng Zhang, **Lawrence Liu**, et al. , “Synchronization network-based approach for accurate epileptogenic zone identification from short interictal EEG data ”, To be submitted to *Nature Neuroscience*, 2025.

TEACHING EXPERIENCE

Teaching Assistant

April 2024 – June 2024

UCLA Electrical and Computer Engineering Department

Los Angeles, CA

- TAed for a graduate class, Deep Learning 2 (ECE 239AS) for spring quarter 2024, lead office hours and discussions
- Created from scratch two Deep Reinforcement Learning projects in python with Deep Deterministic Policy Gradient (DDPG) and Deep Q Learning (DQN)

PROJECTS

Kernel-Density-Estimation Stock Trading Algorithm

April 2023 - June 2023

- Single handily developed a novel Kernel-Density based algorithm to identify optimal portfolio weights for a buy and hold strategy in stocks
- Lead a team of 4 undergraduates representing UCLA against 40+ teams from Stanford, Harvard, CMU, etc at the UChicago 2023 Trading Competition
- Optimized Kernel Parameters with Stochastic Gradient Descent for greater accuracy, efficiency, and scalability
- Utilized Spectral Clustering to identify sectors of correlated stocks, iterative applied this algorithm on each sector
- Backtested on 2 years of real world S&P 500 data and was able to achieve a 17.4% year over year return and a Sharpe ratio of 1.424, outperformed the S&P500 and naive Markowitz Baseline

TECHNICAL SKILLS

Languages: Python, C/C++, R, Matlab, Octave, Julia, Mathematica

Tools: Git, Bash Scripting, Pytorch, Sklearn, Tensorflow, Pandas, NumPy, Matplotlib, Simulink, Gymnasium, Optuna, Cvxpy, Mosek, Cuda, Transformers

Techniques: Classical Machine Learning, Deep Learning, Reinforcement Learning, Graph Neural Networks, Time Series Analysis, Signal Processing, Data Mining, Data Visualization, Statistical Analysis, Optimization, Numerical Analysis, Control Theory, Model Parallelism & Scaling, Generative AI